AI for Tables

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Overview

Tables

Table Analysis

Table Synthesis

Demo

Tables

Tables

Weather		
Date	Temp.	Humid.
1/1/2020	20°C	58%
1/2/2020	21°C	55%
1/3/2020	22°C	53%

Sales		
Product	Qty.	Price
Milk	1	$3.00
Bread	2	$2.00
Apple	3	$1.00

Clients		
Name	Age	Country
John Doe	21	USA
Zhang San	28	CHN
Priya Singh	39	IND

Tabular Data

Weather			
Date	Temp.	Humidity	Wind
1/1/2020	20°C	58%	14 km/h
1/2/2020	21°C	55%	15 km/h
1/3/2020	22°C	53%	17 km/h

Tabular Data

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Tabular Data

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1/3/2020	22°C	53%	17 km/h

Types of Tabular Data

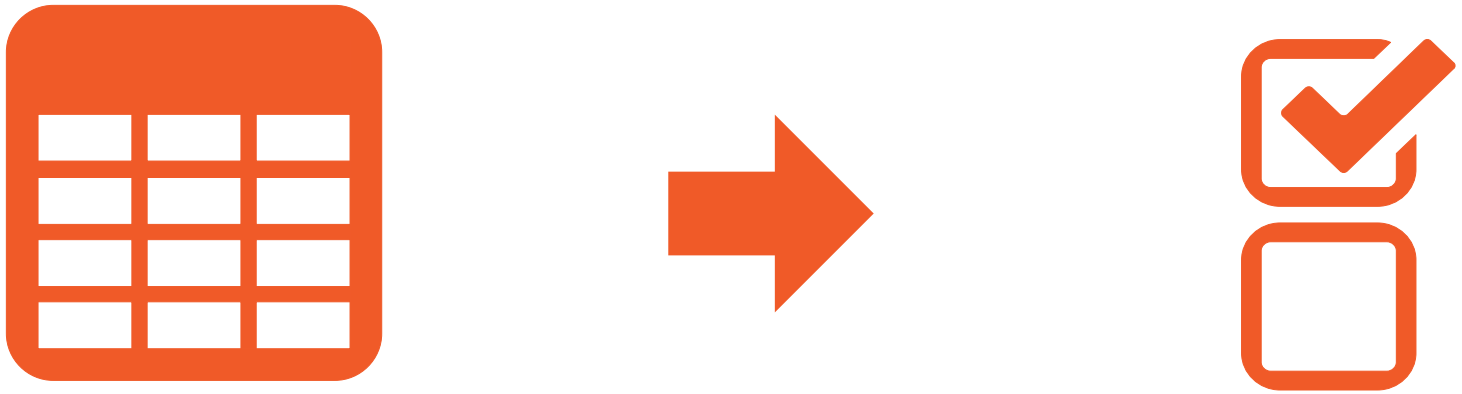
Categorical	Numerical
true, false	1, 2, 3 ...
red, green, blue	4.56
small, medium, large	5.97×10^{24}

Table Analysis

Data Analysis Tools

Classification	Regression	Recommendation

Classification



Classification

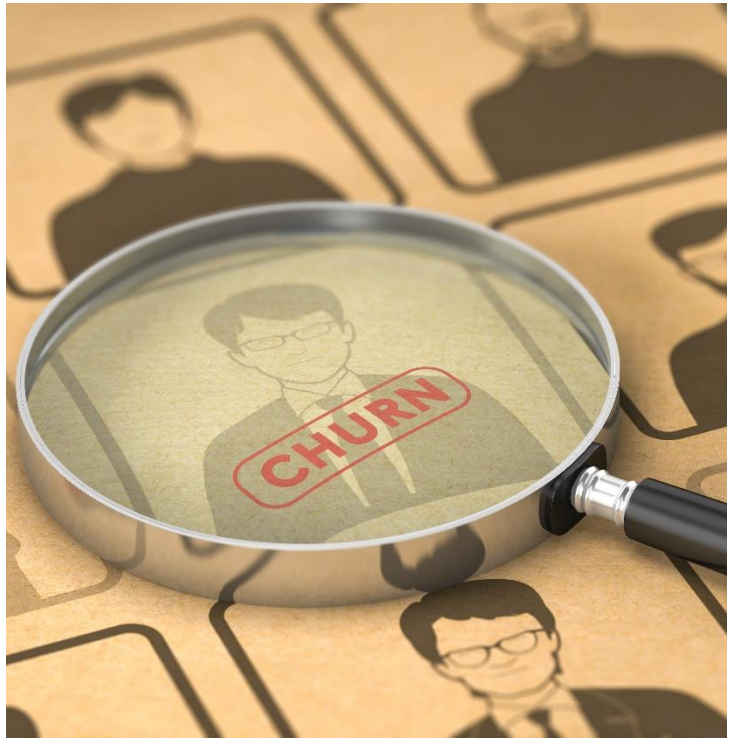
Customers		
Customer	Income	Late Pay
Zhang San	$99,000	0
Priya Singh	$50,000	1
John Doe	$25,000	3

Output	
Approve?	Score
Yes	90%
Yes	60%
No	80%

Classification

Loan approval

Fraud detection



Customer Churn

Regression

1.23

Regression

Houses		
Area	Bedrooms	Bathrooms
4,000	4	3
2,000	2	2
1,000	1	1

Output	
Price	Score
$750,000	70%
$500,000	90%
$250,000	80%

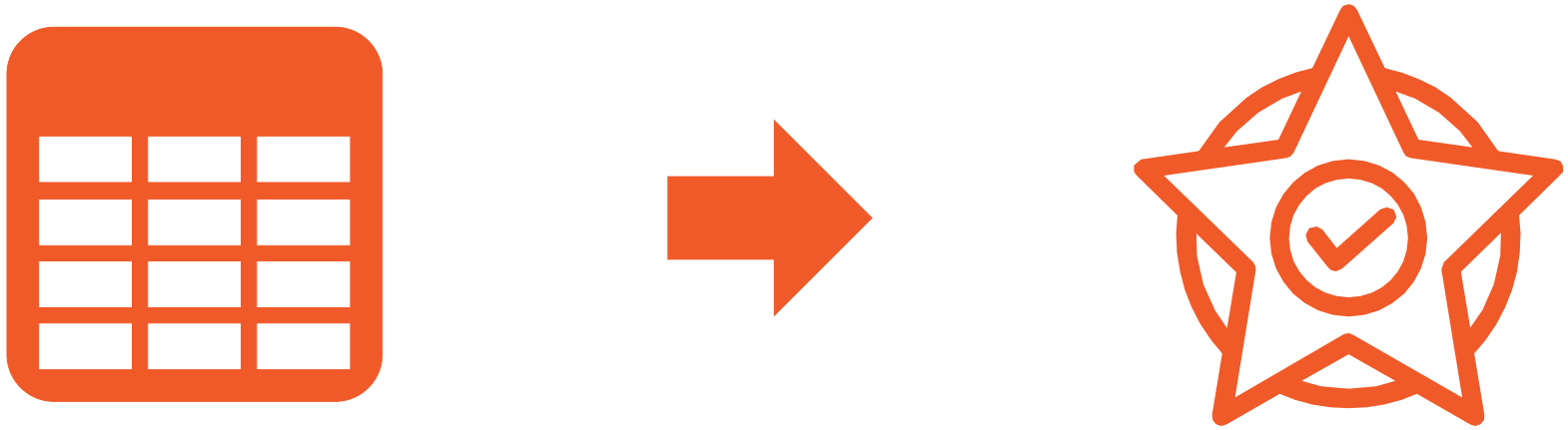
Regression

Sales forecasting

Revenue estimation

Patient outcomes

Recommendation

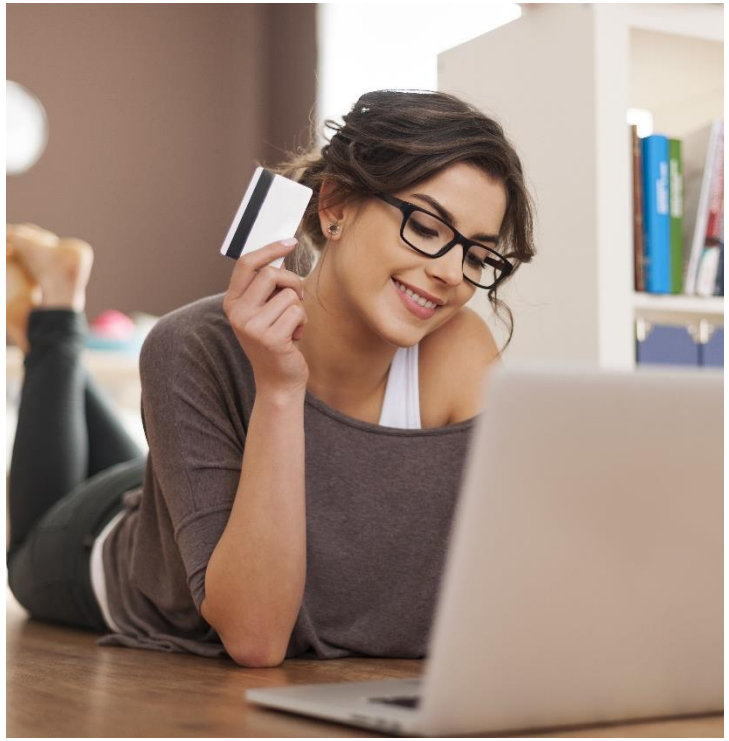


Recommendation

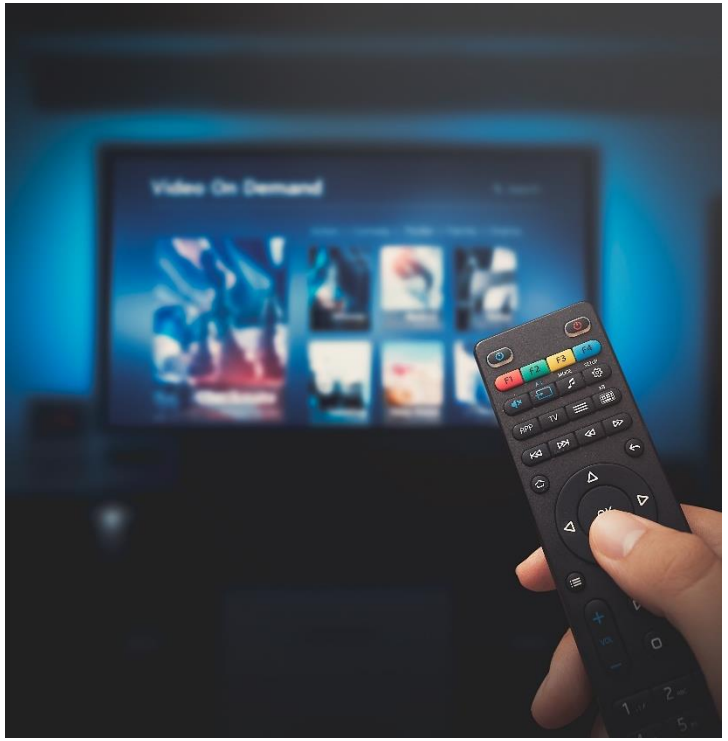
Purchases	
Product	Quantity
Chips	1
Milk	2
Cheese	3

Output	
Recommend	Score
Salsa	70%
Cookies	90%
Crackers	80%

Recommendation



Product upselling



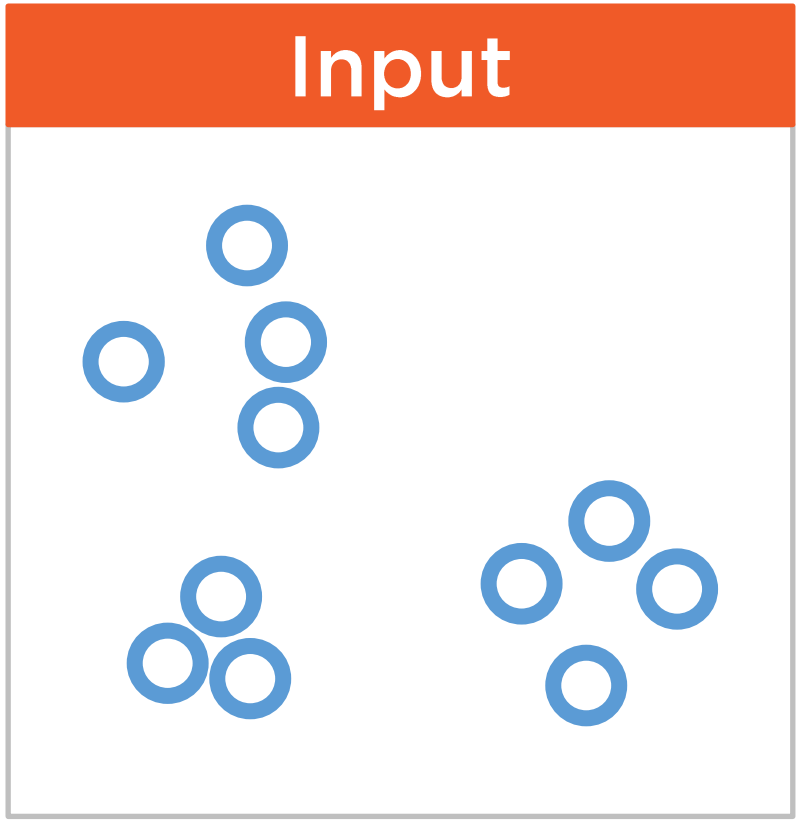
Video recommendation

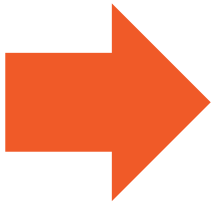
Friend suggestions

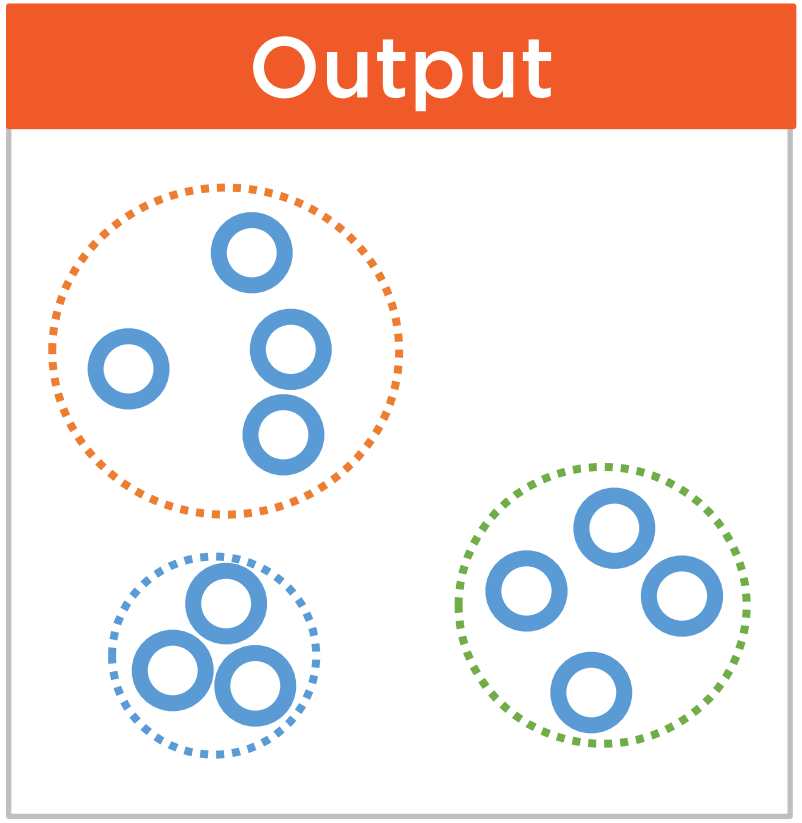
Other Table Analysis Tools

Classification	Regression	Recommendation
Clustering	Association	Anomaly detection

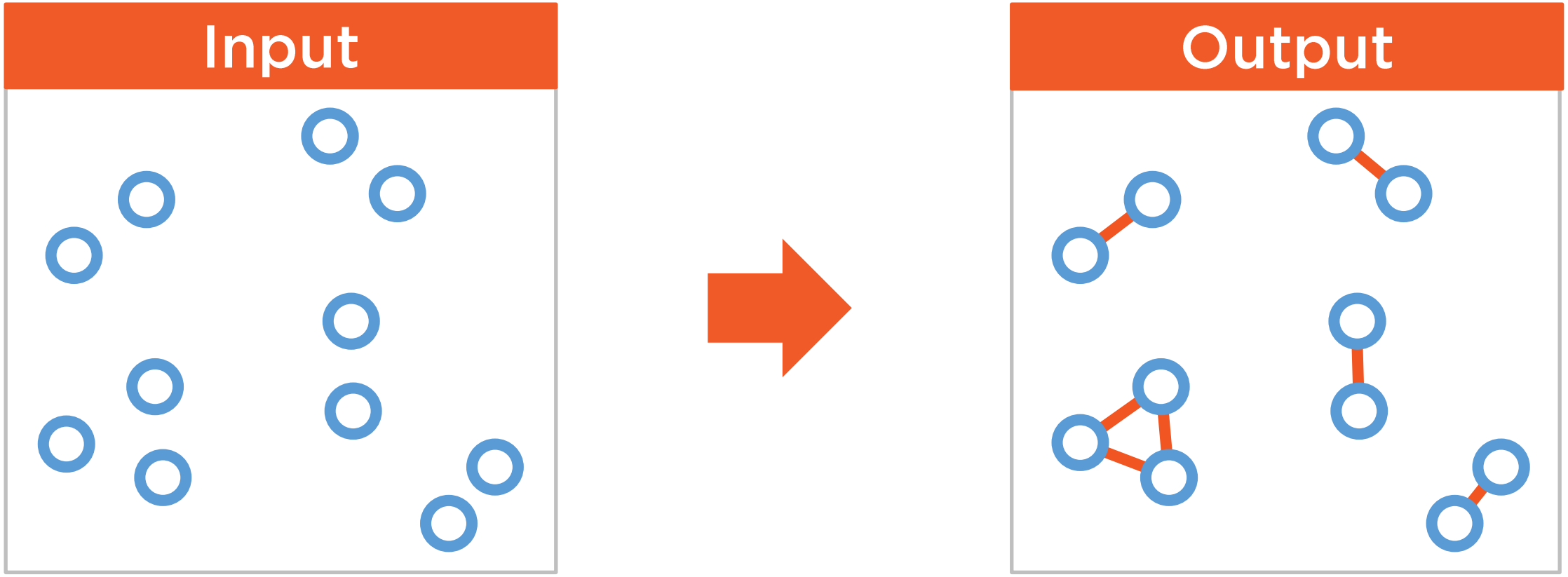
Clustering







Association



Anomaly Detection

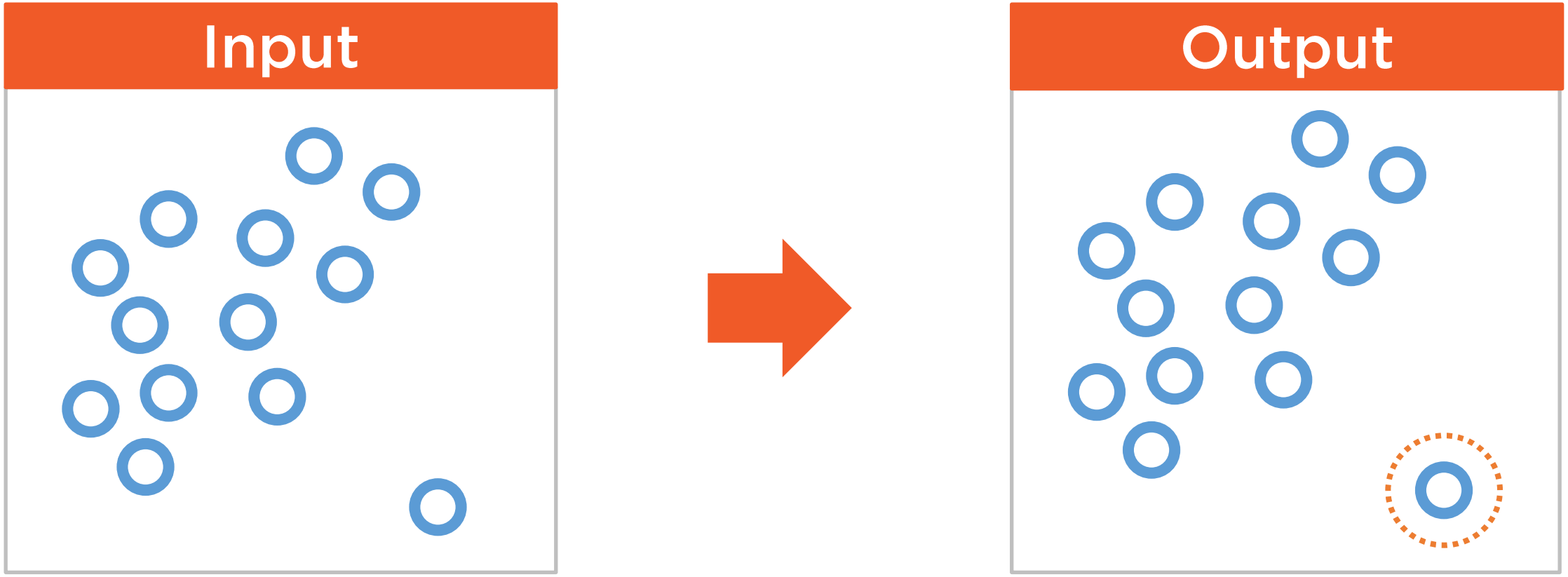


Table Synthesis

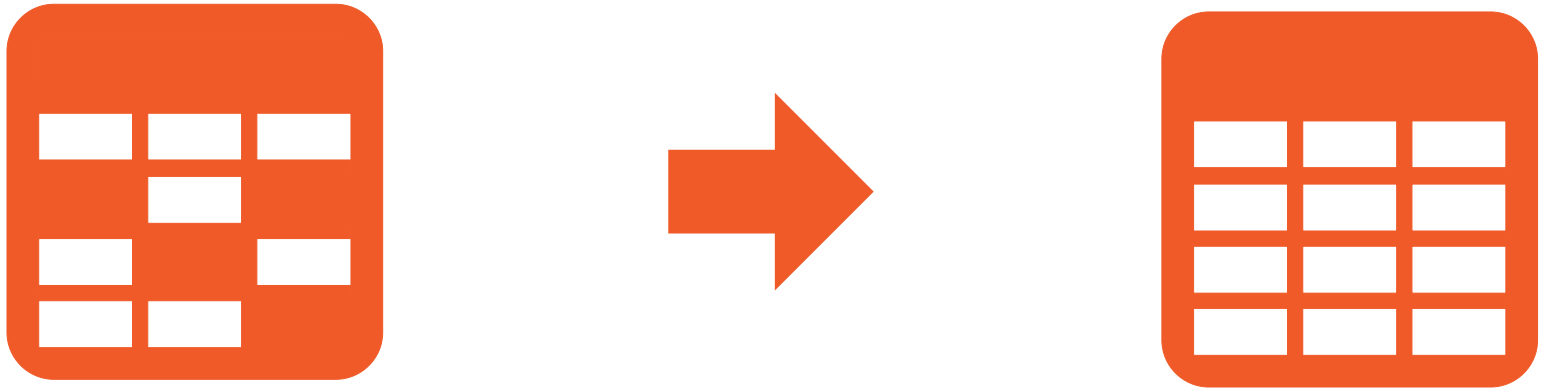
Table Synthesis Tools

Data imputation

Data generation

Data transformation

Imputation



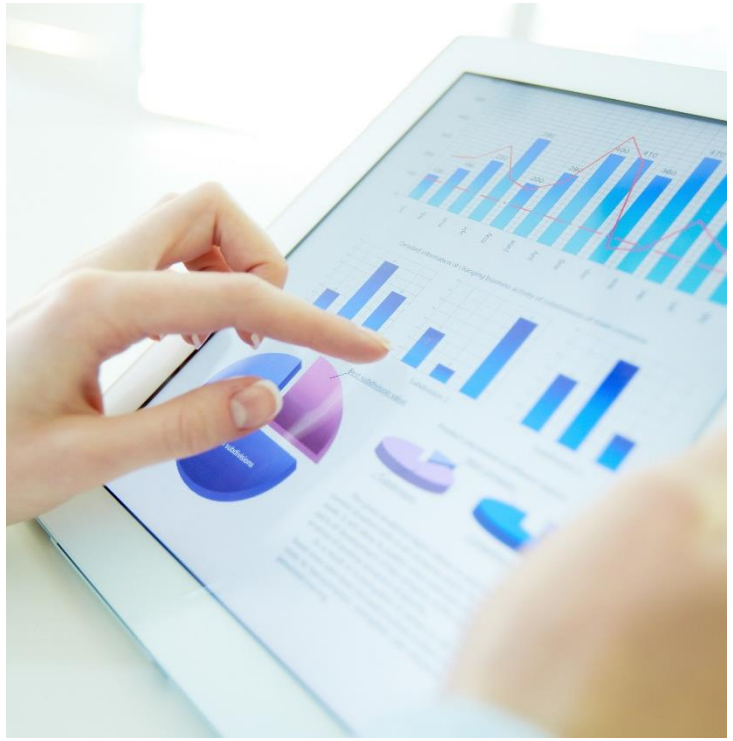
Imputation

Weather		
Date	Temp.	Humid.
1/1/2020	[null]	58%
[null]	21°C	55%
1/3/2020	22°C	[null]

Weather		
Date	Temp.	Humid.
1/1/2020	20°C	58%
1/2/2020	21°C	55%
1/3/2020	22°C	53%

Imputation

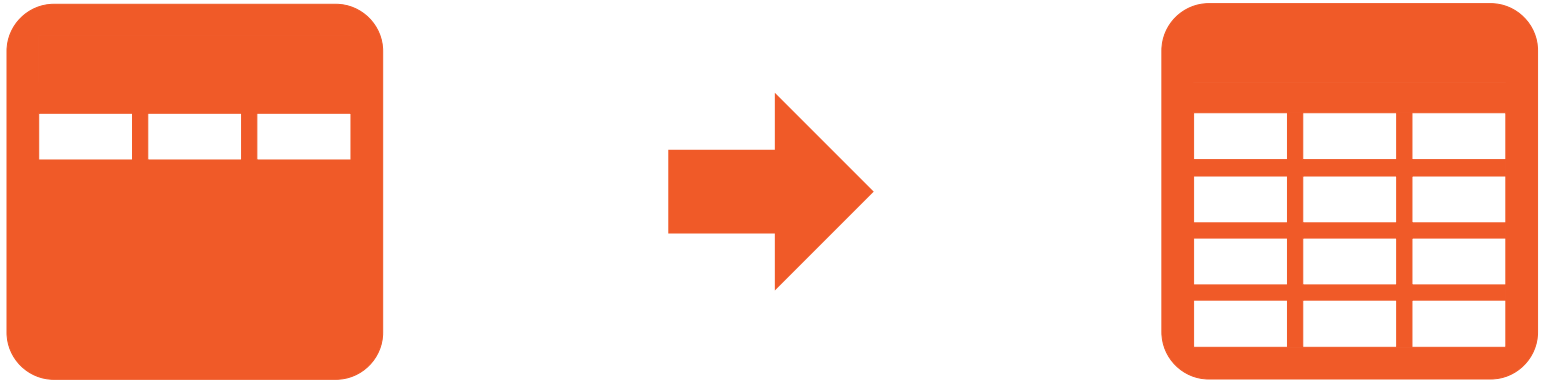
Data visualization



Aggregate analysis

ML training datasets

Generation



Generation

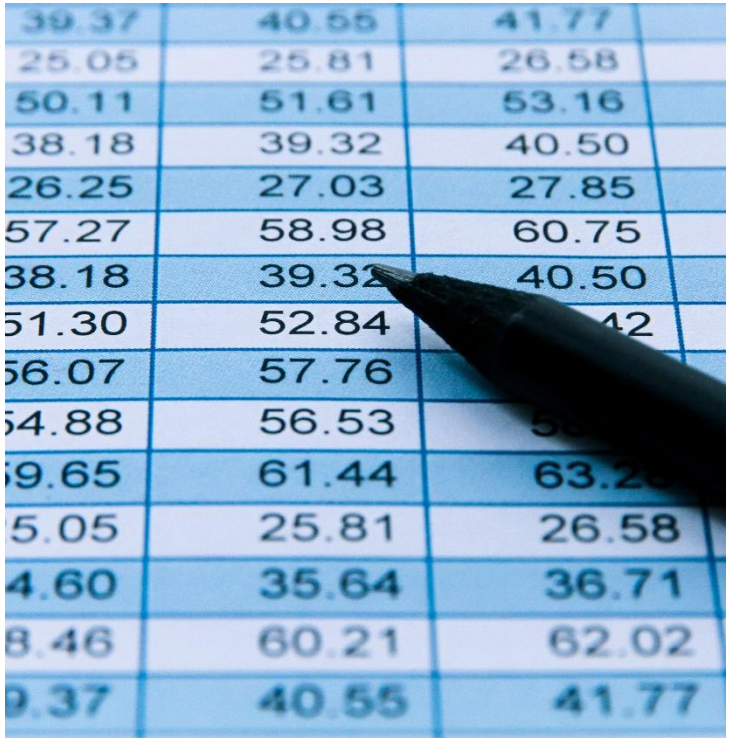
Customers		
Customer	Income	Late
Zhang San	$99,000	0
Priya Singh	$50,000	1
John Doe	$25,000	5

Customers		
Customer	Income	Late
Kim Li	$90,000	0
Mike Ray	$45,000	2
Jose Gil	$30,000	4

Generation

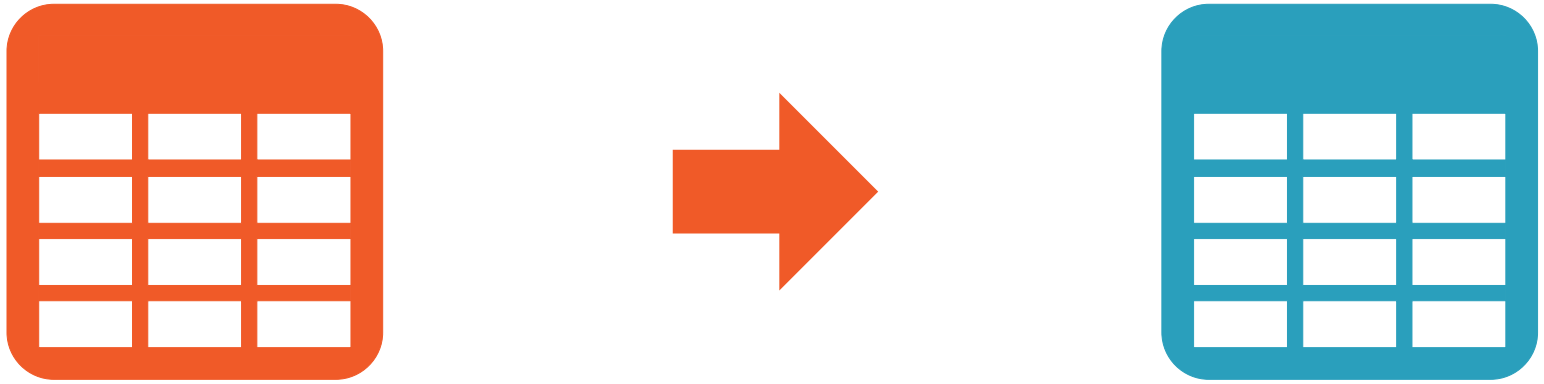
Application testing

Data anonymization



Data augmentation

Transformation

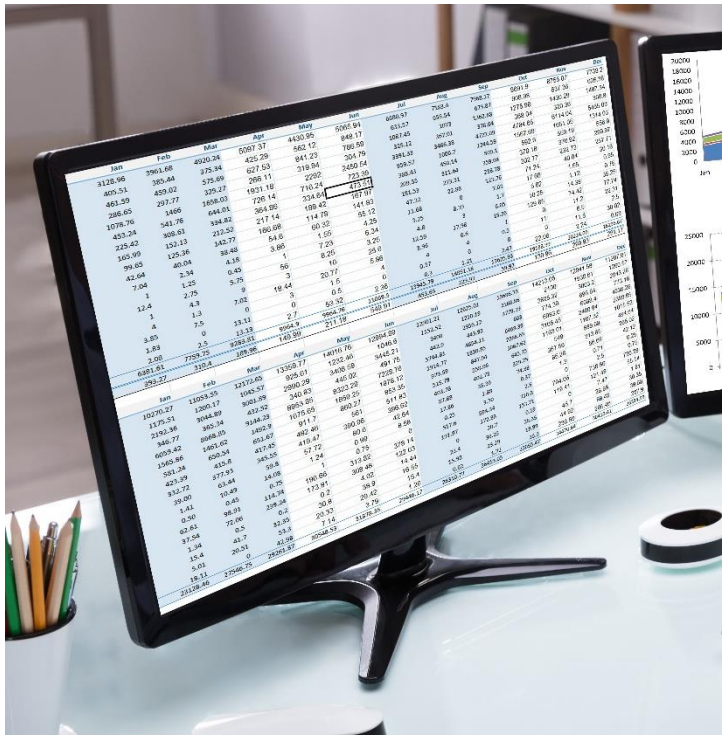


Transformation

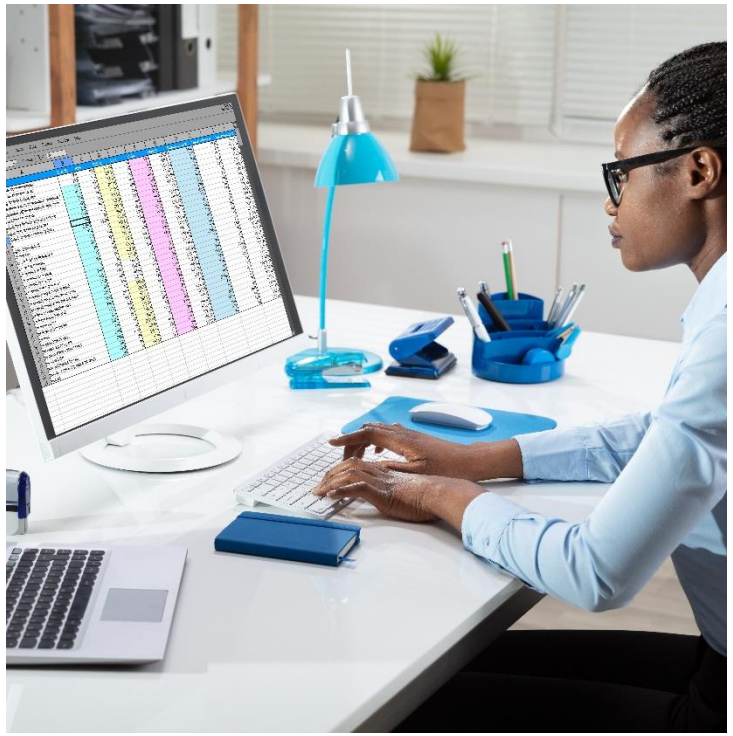
Weather	
Date	Temp. Range
Jan 1, 2020	15°C - 25°C
Jan 2, 2020	16°C - 26°C
Jan 3, 2020	17°C - 27°C

Weather		
Date	Low	High
1/1/2020	15°C	25°C
1/2/2020	16°C	26°C
1/3/2020	17°C	27°C

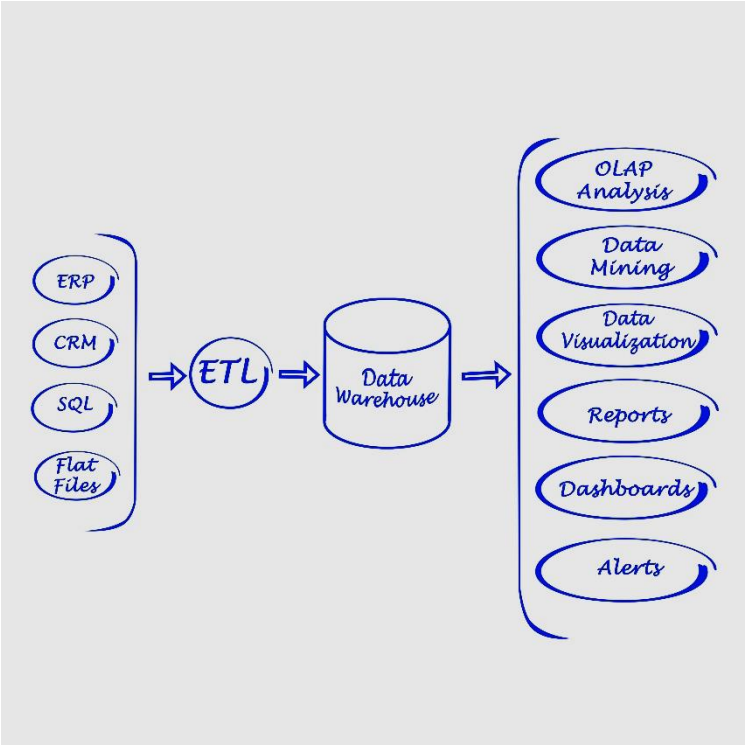
Transformation



Spreadsheets



Ad-hoc analysis



Data pipelines

Tabular Data Classification Demo

Titanic Survivor Dataset

Sex	Age	Family	Class	Fare	Cabin	Port	Survived
female	29	0	1	$211.33	B5	South…	Yes
male	0.9	3	1	$151.55	C22	South…	Yes
female	2	3	1	$151.55	C22	South…	No
male	30	3	1	$151.55	C22	South…	No
female	25	3	1	$151.55	C22	South…	No
male	48	0	1	$26.55	E12	South…	Yes
…	…	…	…	…	…	…	…

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Input

Output

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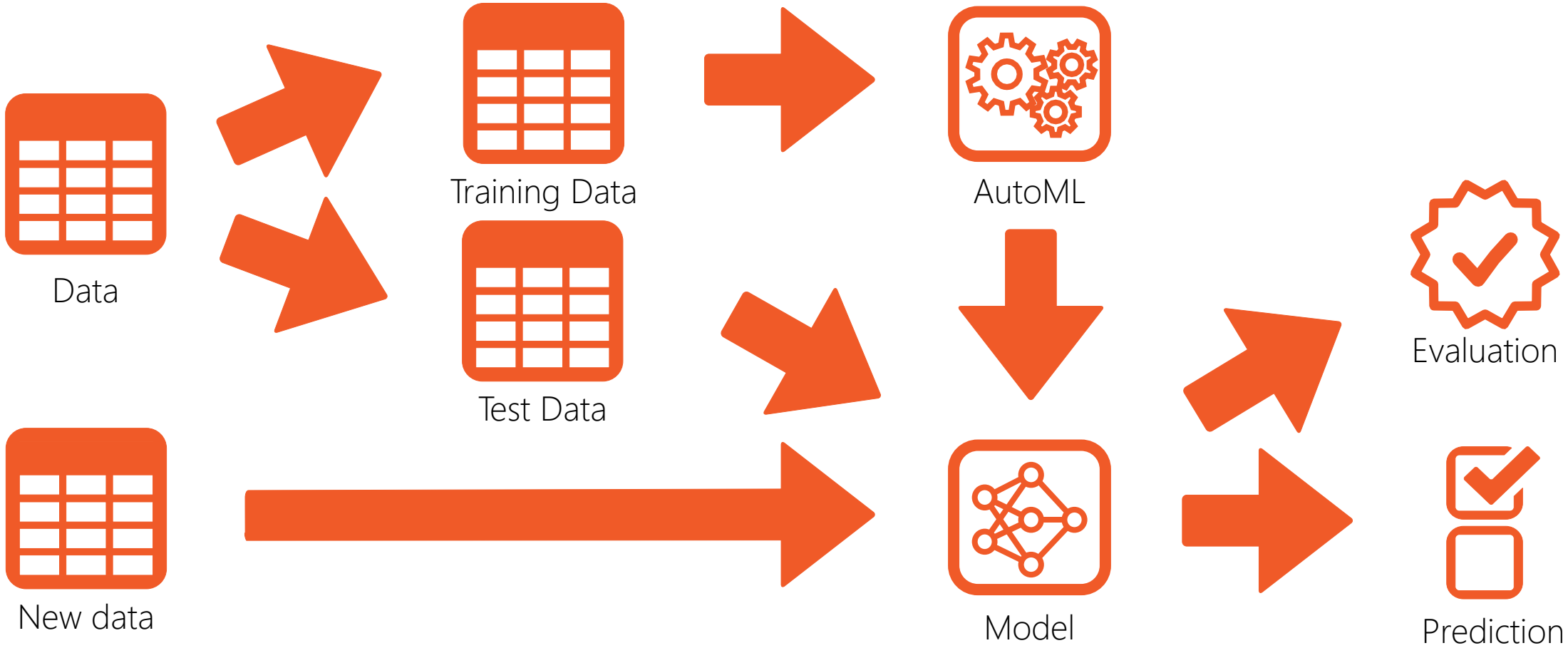
Demo Overview



Instructions

https://github.com/matthewrenze/ai-developers-toolkit/

Summary

Tables

Table Analysis

Table Synthesis

Demo